



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

First Semester M.Tech. (CBS) (CBCS) Degree Examinations

December 2024 – January 2025

22CBS111 – CLOUD SECURITY AND IOT SECURITY

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each
and **One full** question from **Unit – III**.

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	With the help of a neat diagram, explore virtualization hardware. Explain the advantages of cloud virtualization.	10	L*2	1	1,2
	b)	What is a hypervisor? Compare and contrast the two types of hypervisors and discuss their significance.	10	L2	1	1
2.	a)	What is cloud computing? Why is cloud computing important? What are the key benefits offered by cloud computing?	10	L2	1	1
	b)	Discuss the advantages and disadvantages of public, private, and hybrid cloud deployment models.	10	L2	1	1,2
3.	a)	What is the shared responsibility model? Explore the respective responsibilities of the customer and the cloud service provider in AWS framework.	10	L2	1	1
	b)	What is threat modeling? What are the different phases of the threat modeling process? Illustrate the different phases of threat modelling with a real-time example.	10	L2	1	1
		Unit – II				
4.	a)	Enumerate the different phases of Penetration Testing. How do Vulnerability Scanning and Penetration Testing differ?	10	L2	2	1
	b)	Illustrate how key management functions with a clear diagram.	10	L2	2	1,2
5.	a)	Identify and explain the key concepts for ensuring data security in the cloud.	10	L2	2	1,2
	b)	List the various types of attacks. Provide a detailed explanation of SQL injection with the help of a diagram.	10	L2	2	1
6.	a)	Describe the key components of cloud security and discuss the associated benefits. What are the major advantages and challenges faced in cloud security?	10	L2	2	1,2
	b)	With the help of the diagram explain the role of Identity Management (IDM) system in online shopping system.	10	L2	2	2
		Unit – III				
7.	a)	With the help of a diagram, explain the IoT Reference Model and its significance.	10	L2	4,5	1
	b)	Explain the specific security threats associated with the Sensing/Perception Layer in IoT.	10	L2	4,5	1,2
8.	a)	Outline the essential security requirements in the Internet of Things (IoT) ecosystem.	10	L2	4,5	1
	b)	List and discuss the key security concerns within the IoT four-layer architecture.	10	L2	4,5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
